

DS	Session	Hashing	Question Information	Level 1	Challenge 99
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Problem Description

There are N chocolates denoted by array A where $A[i]$ is the length of the i -th chocolate.

Alice can melt each chocolate and then convert it into a chocolate whose length is any divisor of the number $A[i]$.

So, a chocolate of length $A[i]$ can be converted into X different types of chocolate where X is the count of divisors of the number $A[i]$.

So you need to count the total unordered pair of chocolates such that their X value is same.

Constraints

$1 \leq N \leq 10^5$
 $1 \leq A[i] \leq 10^6$

Input Format

The first line contains an integer N as input denoting the total number of elements in the array A .

The next line contains N space-separated integers that denote the elements of the array A .

Output Format

In the output, print the total number of ways as mentioned in the statement.

▼ Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN) 3 2 3 4 EXPECTED OUTPUT 1	INPUT (STDIN) 10 2 3 4 2 7 6 8 6 4 9 EXPECTED OUTPUT 12
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✓ Mandatory Test Cases

Test Case 1
KEYWORD
<code>while(--N)</code>

✓ Complexity Test Cases

Test Case 1
CYCLOMATIC COMPLEXITY
8

Test Case 2
TOKEN COUNT
345

Test Case 3
NLOC
44

```
Hashing9.c +
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX_VAL 1000005
5
6 int count_divisors(int n) {
7     int count = 0;
8     for (int i = 1; i * i <= n; i++) {
9         if (n % i == 0) {
10             if (i * i == n) count++;
11             else count += 2;
12         }
13     }
14     return count;
15}
16
17int main() {
18     int N;
19     if (scanf("%d", &N) != 1) return 0;
20
21     int original_N = N;
22     int *A = (int *)malloc(original_N * sizeof(int));
23     for (int i = 0; i < original_N; i++) {
24
25     }
```

```
26
27     int freq_div[1000] = {0};
28     int i = 0;
29     freq_div[count_divisors(A[i++])]++;
30     while(--N) {
31         freq_div[count_divisors(A[i++])]++;
32     }
33
34     long long total_pairs = 0;
35     for (int d = 0; d < 1000; d++) {
36         if (freq_div[d] > 1) {
```

```
37         total_pairs += (long long)freq_div[d] * (freq_div[d] - 1) / 2;
38     }
39 }
40
41 printf("%lld\n", total_pairs);
42
43 free(A);
44 return 0;
45}
46
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

3

2 3 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Hashing9.exe

1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

10

2 3 4 2 7 6 8 6 4 9

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Hashing9.exe

12

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>